TON DUC THANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY
DIVISION OF COMPUTER SCIENCE
SOCIALIST REPUBLIC OF VIETNAM
Independence – Freedom – Happiness

Ho chi minh city, October 31, 2022

COURSE SYLLABUS
INTRODUCTION TO LOGICAL THINKING
COURSE CODE: 503116

1. General information:

Credits	3(3,0)					
Time allocation	**Theory (periods):**	45	**Practice (periods):**	0	**Self-Study (hours):**	90
Prerequisite	No				**Prerequisite Code**	No
Prior-Completion	No				**Prior-Completion Code**	No
Co-requisite	No				**Co-Requisite Code**	No
Program	2 programmes				**Programme Code**	F7480103, F7480101

2. Course objectives(COs):

No.	Course Objectives (COs)
1	Remember basic discrete structures.
2	Understand proof techniques, algorithms related to discrete structures.
3	Apply discrete algorithms to solve problems.
4	Analyze discrete structures and algorithms to solve a specific problem.
5	Evaluate specific discrete structures for a practical problem.

3. Course learning outcomes (CLOs):
After completing this course/module, learners will be able to:

No.	CLOs	F7480103	F7480101
1	Recognize basic discrete structures.	PLO7	PLO7
2	Explain proof techniques, algorithms related to discrete structures.	PLO7	PLO7

No.	CLOs	F7480103	F7480101
3	Use discrete algorithms to solve professional tasks.	PLO4	PLO5
4	Organize discrete structures and algorithms to solve a specific problem.	PLO4	
5	Select specific discrete structures for a practical problem.	PLO5	

4. Brief course content:

Many areas of computer science require the ability to work with the notions from the area of discrete mathematical structures. The goal of the course is to introduce such fundamental material for computer science. The students will obtain knowledge on discrete mathematical structures and techniques which are applied in computing practice. Three kinds of topics are covered, namely basic logic, naïve set theory and proof techniques. The course not only introduces theoretical concepts, but is also oriented to practical applications. For instance, the ability to create and understand a proof, either a formal one or a less formal but still rigorous argument, is important in almost every area of computer science.

5. Student's tasks:
The highest professional standards are expected of all students who study this course. The collective class reputation and the value of the program's experience hinges on this. This includes:

- Being on-time and in your seat when class starts
- Staying until the end of class. Should students have to leave class early, please have the courtesy of letting the instructor know before the beginning of the period and leave quietly so as not to disturb the other members of the class.
- Being constructive to the learning environment. Mobilephones (both smart and inept) and other electronic devices need to be set to silent during class at all times.
- Being "present" in the class and focused on class material. The lectures will sometimes cover extra material (e.g., exercises, discussions) not contained in lecture notes. Students are responsible for everything covered or assigned in class. If students miss a class, it is entirely their responsibility to determine what they have missed including any administrative announcements lecturer may have made.
- Coming prepared to contribute to the class.
- Communication with the lecturer.

Students have to attend at least 80% of class hours (if this regulation is not fulfilled, students are not eligible to take their final exam).

6. Course materials:
- Textbook:
[1]. Susanna S. Epp, [2011], Discrete Mathematics with applications, 4th edition, Brooks/Cole, Boston.

- Supplementary Readings:
[2]. Kenneth H. Rosen, [2012], Discrete mathematics and its applications, 7th edition, McGraw-Hill, New York.

[3]. Allan Stavely, [2014], Programming and mathematical thinking: A gentle introduction to discrete math featuring Python, The New Mexico Tech Press, Socorro, New Mexico.

- Additional readings:
[4]. R. P. Grimaldi, [2004], Discrete and Combinatorial Mathematics: An Applied Introduction, 5th edition, Pearson Education, Boston.

[5]. Rowan Garnier, John Taylor, [2002], Discrete Mathematics for New Technology, 2nd edition, Institute of Physics, Bristol, Philadelphia.

7. Description of assessment:

Assessment categories	Weight (%)	Types of questions	CLO1	CLO2	CLO3	CLO4	CLO5
Process evaluation 1	10	Process Exercise	X	X			
Process evaluation 2	20	Process Exercise	X	X			
Mid-term test	20	Presentation			X	X	X
Final examination	50	Report			X	X	X

8. Schedule:

Session	Content	Organization of teaching				Self-Study	CLOs	Prerequisite-related content	Student's tasks
		T	E	P	D				
	Chapter 1: Logic of Compound Statements	**6**	**3**	**0**	**0**	**18**			
1 - 3	- Logical Form - Logical Equivalence - Conditional Statements - Valid Arguments - Invalid Arguments **Teaching methods:** - Presentation - Discussion - Exercise	6	3	0	0	18	[1], [2], [3], [4], [5]		**At class:** - Positive listening, asking questions. - Do the exercises required by the instructor. **After class:** - Read the lecture before class - Do homework. - Read [1], chapter 2

Session	Content	Organization of teaching				Self-Study	CLOs	Prerequisite-related content	Student's tasks
		T	E	P	D				
	Chapter 2: Logic of Quantified Statements	**6**	**3**	**0**	**0**	**18**			
4 - 6	- Predicates - Quantifiers - Quantified Statement - Statements with Multiple Quantifiers - Arguments with Quantified Statements **Teaching methods:** - Presentation - Discussion - Exercise	6	3	0	0	18	[1], [2], [3], [4], [5]		**At class:** - Positive listening, asking questions. - Do the exercises required by the instructor. **After class:** - Read the lecture before class - Do homework. - Read [1], chapter 3
	Chapter 3: Proof techniques	**3**	**3**	**0**	**0**	**12**			
7 - 8	- Proof by Construction: Existence Proof, Disproof by Counterexample - If-then Statements - For-all Statements - Proof by Contradiction - Proof by Controposition **Teaching methods:** - Presentation - Discussion - Exercise	3	3	0	0	12	[1], [2], [3], [4], [5]		**At class:** - Positive listening, asking questions. - Do the exercises required by the instructor. **After class:** - Read the lecture before class - Do homework. - Read [1], chapter 4
	Chapter 4: Sequence and recursion	**2**	**1**	**0**	**0**	**6**			
9	- Sequences: Explicit formula, recurrence relation - Common sequences - Solving recurrences - Application: Tower of Hanoi **Teaching methods:** - Presentation - Discussion - Exercise	2	1	0	0	6	[1], [2], [3], [4], [5]		**At class:** - Positive listening, asking questions. - Do the exercises required by the instructor. **After class:** - Read the lecture before class - Do homework. - Read [1], chapter 5
	Chapter 5: Set and Relation	**3**	**3**	**0**	**0**	**12**			
10 - 11	- Basic Set Theory - Operation on Sets - Cartesian product - N-ary relation - Properties of Relations on a Set - Equivalence Relations - Transitive Closure - Partial and Total Orders **Teaching methods:** - Presentation - Discussion - Exercise	3	3	0	0	12	[1], [2], [3], [4], [5]		**At class:** - Positive listening, asking questions. - Do the exercises required by the instructor. **After class:** - Read the lecture before class - Do homework. - Read [1], chapter 6, 8
	Chapter 6: Functions	**2**	**1**	**0**	**0**	**6**			
12	-Definition -Properties -Composition -Exercises **Teaching methods:** - Presentation - Discussion - Exercise	2	1	0	0	6	[1], [2], [3], [4], [5]		**At class:** - Positive listening, asking questions. - Do the exercises required by the instructor. **After class:** - Read the lecture before class - Do homework. - Read [1], chapter 7
	Chapter 7: Cardinality	**3**	**3**	**0**	**0**	**12**			

Session	Content	Organization of teaching				Self-Study	CLOs	Prerequisite-related content	Student's tasks
		T	E	P	D				
13 - 14	-Introduction -Cardinality -Infinity -Beyond Infinity **Teaching methods:** - Presentation - Discussion - Exercise	3	3	0	0	12	[1], [2], [3], [4], [5]		**At class:** - Positive listening, asking questions. - Do the exercises required by the instructor. **After class:** - Read the lecture before class - Do homework. - Read [1], chapter 7
	Chapter 8: Review	**0**	**3**	**0**	**0**	**6**			
15	Review **Teaching methods:** - Presentation - Discussion - Exercise	0	3	0	0	6	[1], [2], [3], [4], [5]		**At class:** - Positive listening, asking questions. - Do the exercises required by the instructor. **After class:** - Read the lecture before class - Do homework.
	Total	**25**	**20**	**0**	**0**	**90**			

9. Date of first approval:

October 31, 2022

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10. Date of update:

No version has been issued yet.